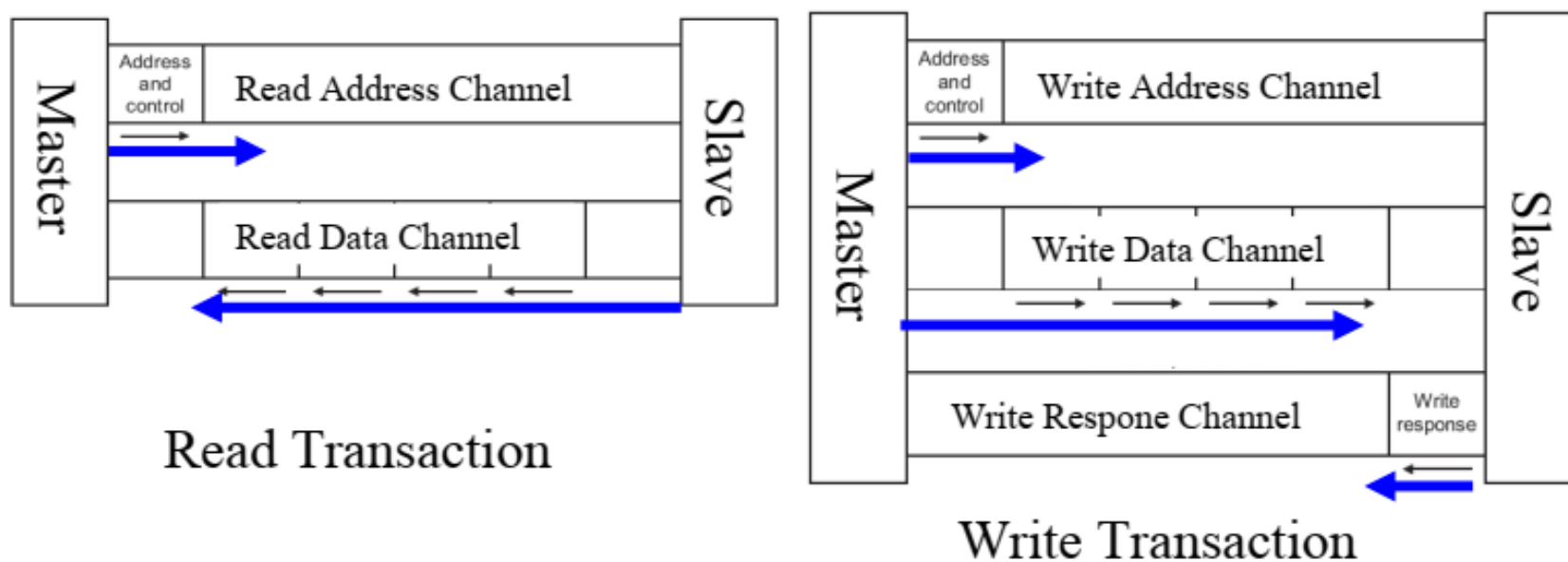


AMBA Specification
Advanced eXtensible Interface Bus
(AXI)

Architecture

- ❖ **AXI protocol is Burst-based** transactions with only start address issued.
- ❖ **Address/Control is issued ahead of actual data transfer.**
- ❖ **5 channels.**



Key Features of AXI Protocol

- ❖ Separate **address/control** and **data** phases
- ❖ Separate **Read** and **Write** data channels
- ❖ Support for **unaligned** data transfers using **byte strobes**
 - ❖ **Ex: Access a 32-bit data that starts at address 0x80004002**
- ❖ **Burst-based** transactions with only start address issued
- ❖ Ability to issue multiple outstanding addresses
 - ❖ **ID signals**
- ❖ Out of order transaction completion
 - ❖ **ID signals**
- ❖ Easy addition of register stages to provide timing closure
 - ❖ **5 channels independent, each channel transfers info. in only one direction.**

Channel Definition

- ❖ **Five independent channels which consists of two-way handshake signals**

 - ❖ **VALID, READY**

- ❖ **VALID**

 - ❖ **Asserts when valid data or control information are available on the channel**

- ❖ **READY**

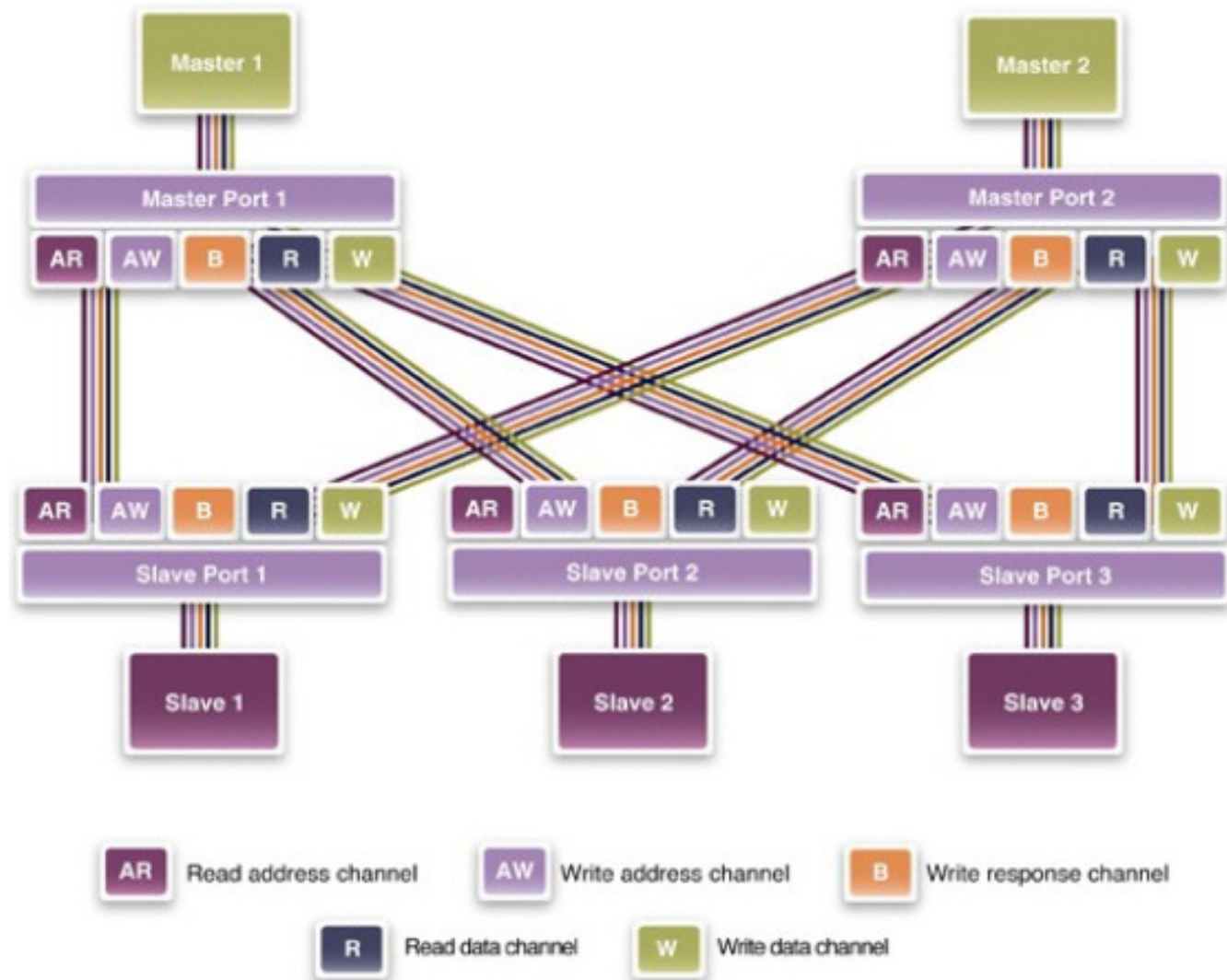
 - ❖ **Asserts when receiver can accept the data**

- ❖ **LAST**

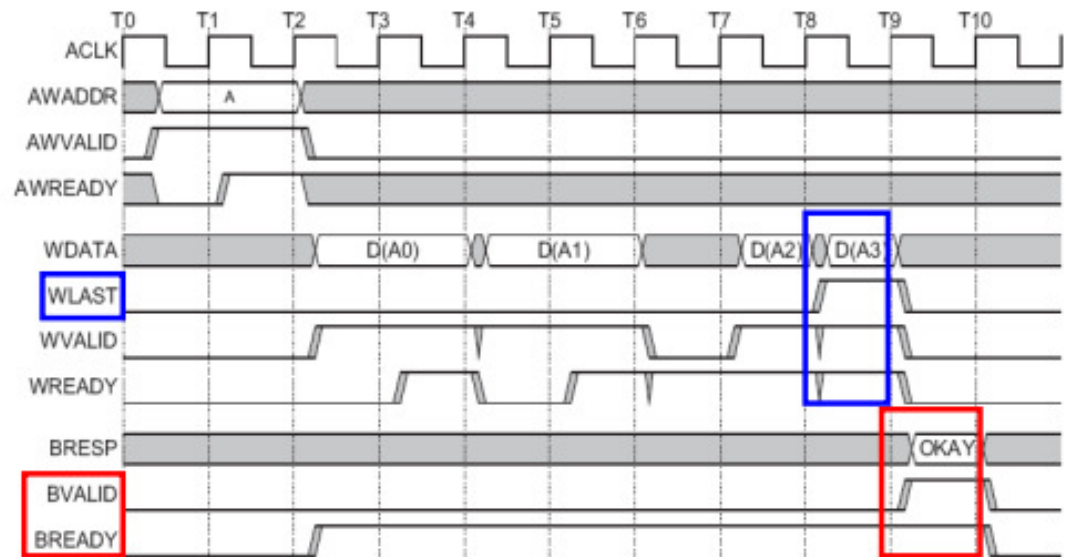
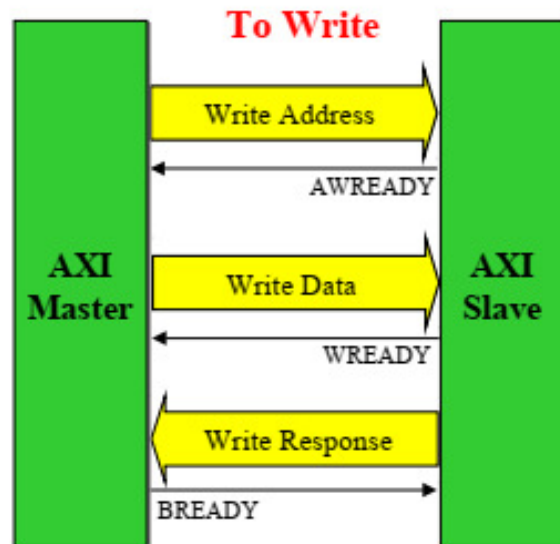
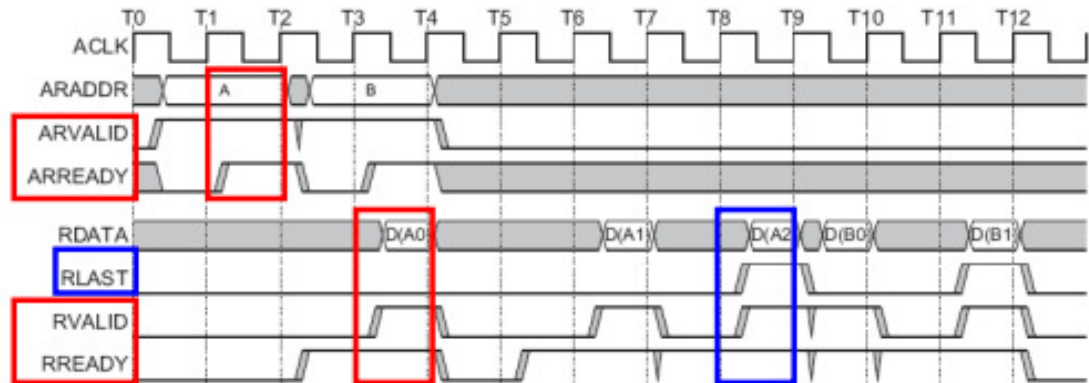
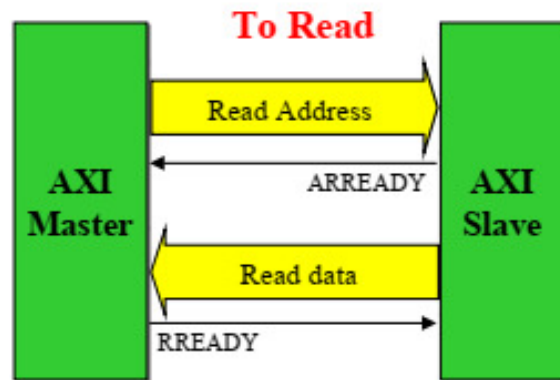
 - ❖ **Asserts while the final data completes**

Flexible Interconnect

- **Shared** address and data buses
- **Shared** address buses and multiple data buses
- **Multilayer**, with multiple address and data buses

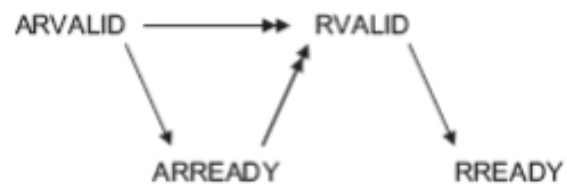


Basic AXI Transactions



AXI Transaction Handshake Dependencies

Read



Write data can appear **before** Write address

Write data appear in **the same cycle** as the address

Read data always come **after** Read address

Write

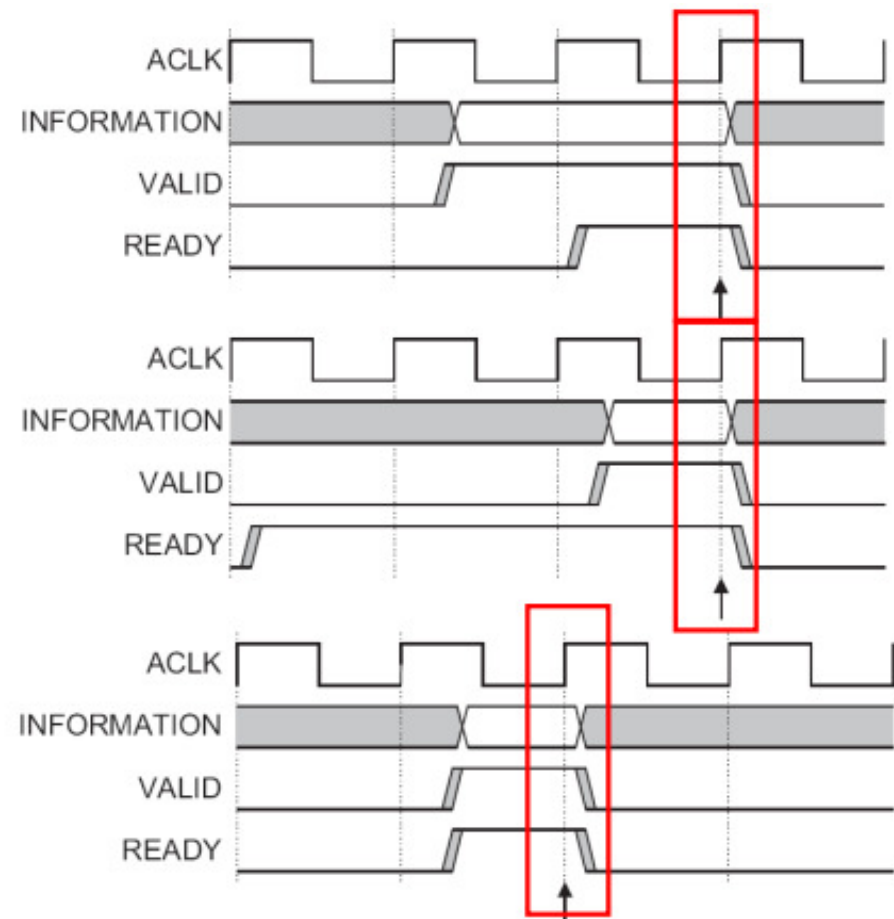


Write response always come **after** write data

AXI Channel Handshaking

The source presents the data /control_info and **drives the VALID signal HIGH**.

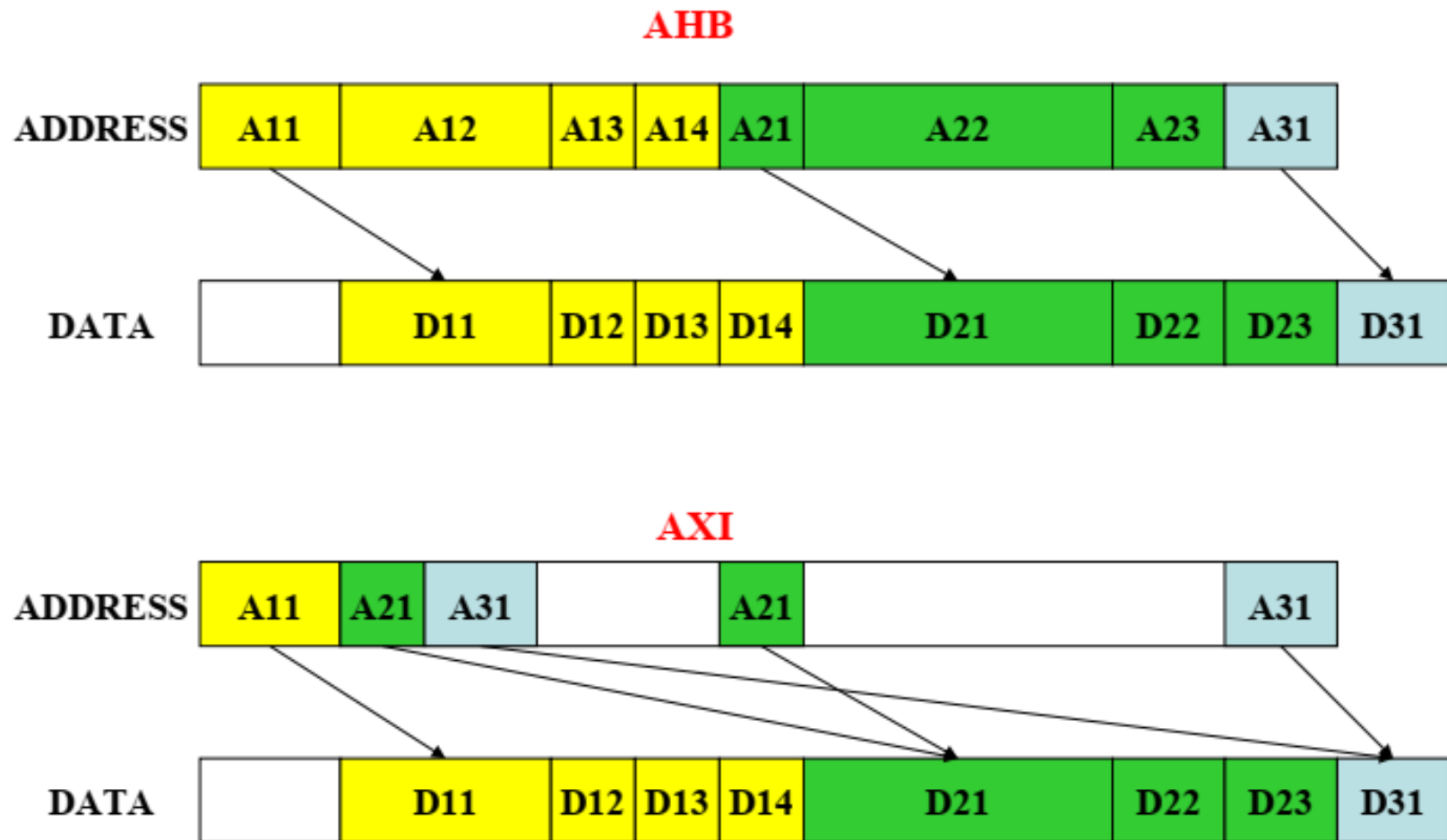
The data/control_info from the source **remains stable** until the destination drives the **READY signal HIGH**, indicating that it accepts the data/control_info.



AXI Signals



AHB/AXI Burst Transfer



AHB/AXI Burst Transfer

